

Quadratic functions - Set 3

TMUA practice questions by topic. Specification MM1.3, M4.5, M4.11-M4.14. 2 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.
- A** 229
B 195
C 4
D 221
E 225
F 255
- 2 p and q are real numbers. Which of the following statements is/are true?
- I** If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .
II If $x^2 + px + q > 0$ for all real x then $q > 0$.
III If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III